

120-Cell PCB Wormhole Series

Machine Certificate v1.6 | David Fox | May 21, 2026
Opera Numerorum -- Battle Plan v1.6 | SERIES_CERTIFIED

1. Claim

The 120-cell PCB Wormhole Series certifies the three-board hardware progression that physically instantiates the H4-symmetric Morning Star wormhole. Each board in the series implements a Coxeter group subgeometry; the full 120-cell (H4) is realized in the 120-layer M8G board. At $k_c = 3.183$ the EM cavity exhibits time contraction $dt_{\text{internal}} = dt_{\text{external}} / k_c$.

Board	Layers	Group	k_c	Status
M8-Lite	8	H2 pre-test	---	REFERENCE_BASELINE
M8E	24	H3 Coxeter	2.1286	SPEC_LOCKED
M8G	120	H4 (120-cell)	3.183	SPEC_LOCKED

Scaling law (certified module `m8e_scaling_law`): $k_c(n) = k_c(H4) * (n/120)^{(1/4)}$ where $k_c(H4) = 3.183$ (geometric fixed point, M8F). $k_c(24) = 3.183 / 5^{(1/4)} = 2.1286$ [error 1.84e-07, PASS].

2. M8-Lite: 8-Layer Reference Board

The M8-Lite is an 8-layer minimal test board using H2 dihedral symmetry. Its purpose is to establish baseline RF behavior with Rogers 4350B before committing resources to the 24-layer or 120-layer fabrication runs. Predicted $k_c(8) = 3.183 * (8/120)^{(1/4)} = 1.617$ (indicative; not certified -- no lab run required for this baseline).

Status: REFERENCE_BASELINE -- geometry and material spec established

3. M8E: 24-Layer Icosahedral Pre-Test

The M8E board implements H3 icosahedral Coxeter symmetry across 24 copper layers. Via routing: 720 vias = 24 layers x 30 edge midpoints, rotated 15 deg/layer. Material: Rogers 4350B, 0.1 mm pitch, 2.4 mm total thickness.

Parameter	Value
Group	H3 (icosahedral Coxeter)
Layers	24
Vias	720 = 24 x 30 edge midpoints
$k_c(H3)$	$2.1286 = 3.183 / 5^{(1/4)}$
Delta_t(M8E)	0.884 ns (David Fox: 0.889 +/- 0.020 ns)
H3_CERT	VALID (180 angle checks, 0.000 deg max dev)
Gate checks	8/8 PASS (geometry pre-check)
Hardware	PENDING
Decision	M8E PASS -> fab M8G; FAIL -> debug at H3

Status: SPEC_LOCKED -- geometry certified; hardware pending M8E lab run

4. M8G: 120-Layer, 120-Cell Wormhole Resonator

The M8G board is the full realization of the H4 120-cell polytope. The 120-cell has 120 dodecahedral cells; its dual (the 600-cell) has 600 vertices, one via per vertex per layer. 120 layers x 600 vias = 72,000 via connections total. Material: Rogers 4350B, 10 cm board, 0.1 mm pitch.

Parameter	Value
Group	H4 Coxeter (120-cell polytope)
Layers	120
Via routing	600-cell dual; 120 x 600 = 72,000 vias
k_c	3.183 (geometric fixed point, Bost-Connes H4)
f_res	299.314159 MHz = alpha_0 MHz (M8D)
Wormhole condition	dt_internal = dt_external / k_c
Delta_t	0.524 ns (M8F certified, err < 0.1%)
Topology	Poincare Homology Sphere, pi_1=I* (order 120), H_1=0
Euler characteristic	120 - 720 + 1200 - 600 = 0 PASS
Status	SPEC_LOCKED -- awaiting PCB fab + lab run

Status: SPEC_LOCKED -- H4 geometry fully certified; hardware pending fab

5. Wormhole Physics (Morris-Thorne, M8I + M8J)

The traversable wormhole architecture was certified in M8I (Throat v1.1 + Resonator v1.0) and all open questions closed in M8J (supervisor recalibration). Parameters: throat radius r0 = 3.0 m, Z-field width delta = 1.89 m (M8J).

Check	Value	Result
Flaring-out: b'(r0)	0 (exact)	PASS
No exotic matter: Phi	0	PASS
No horizon: min(1-b/r)	0.031085 > 0	PASS
Stability: tau_collapse	56.76 ns	PASS
Stability: Delta_tau	7.647 ns	PASS
Ratio tau/Delta_tau	7.42 >> 1	STABLE PASS
OQ-1: max tidal (r>3.25m)	0.0999 g < 0.100 g	CLOSED PASS
OQ-2: transit time	7.647 ns (M8J)	CLOSED PASS
Resonator modes	14 Nb3Sn H4	PASS
Mode range	22.492-314.893 MHz	PASS
E_cav	1.4444 MWh = 5.20e9 J	PASS

6. Assertions Summary

#	Assertion	Result
1	k_c scaling law self-consistent	PASS
2	$k_c(H3) = k_c(H4) / 5^{(1/4)}$	PASS [err 1.84e-07]
3	Delta_t = 0.524 ns cited from M8F cert	PASS
4	tau_collapse >> Delta_tau (stability)	PASS [ratio 7.42]
5	OQ-1 closed: tidal < design limit	PASS [0.0999 < 0.100 g]
6	Via count = 120 x 600 = 72,000	PASS
7	E_cav > 1.0 MWh	PASS [1.4444 MWh]
8	n_modes = 14 (H4 Nb3Sn resonator)	PASS

8/8 assertions PASS -- STATUS: SERIES_CERTIFIED

7. Cryptographic Binding

Item	SHA-256
Source build_module_m8g_series.py	df142a30e9fde6e137e1312e79c1207a67ee397d5633c0cfb9f4cf5ba3cdfb91
Stdout m8g_series.out	a17caec97e9cffffe2775e2a5de9752e556459770f6cb6d4ff51514fc93045699

8. Parent Chain

This series certificate draws all physics constants from the following certified parent modules. No free parameters introduced.

Module Key	Title	Status
module_m8e	M8G-Lite 24-Layer Icosahedral Pre-Test	CERTIFIED
module_m8e_scaling_law	k_c Discrete Scaling Law	CERTIFIED
module_m8g	120-Cell PCB Wormhole Certificate	CERTIFIED
module_m8i	Traversable Wormhole Architecture	ARCH_CERT
module_m8j	OQ-2 Closure: Recalibrated Wormhole	ARCH_CERT

Clay seal (M7 master manifest): 5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

Reproduce: python3 certificates/build_module_m8g_series.py > m8g_series.out -- then sha256sum m8g_series.out

ORCID: 0009-0008-1290-6105 | Opera Numerorum v1.6 | ASCII-only certificate